

4 - States of matter

12/12/24

Pressure

→ if you put a gas into a container, it exerts pressure

→ this is due to collisions between gas molecules and the walls of the container

* temperature $\uparrow$ = Pressure $\uparrow$ particles have higher KE thus collide more

* volume $\uparrow$ = pressure $\downarrow$ particles have more space to move thus collide less



Ideal gases

→ gas molecules move rapidly & randomly

→ volume of the molecules is negligible (considered zero)

→ no intermolecular forces of attraction or repulsion between molecules

→ no kinetic energy is lost in collisions

→ temperature of gas is related to the average KE

* in reality, gases don't perfectly fit this description

ideal gas equation:

$$pV = nRT$$

Annotations:

- pressure in Pa (pointing to p)
- volume in m^3 (pointing to V)
- number of moles (pointing to n)
- gas constant = $8.31 \text{ J/K}\cdot\text{mol}$ (pointing to R)
- temperature in Kelvin (pointing to T)

e.g Calculate the volume occupied by 0.5 mol of CO_2 at a pressure of 150 kPa & a temp. of 19°C

$$p = 150 \text{ kPa} = 150000 \text{ Pa}$$

$$V = ?$$

$$n = 0.5$$

$$R = 8.31$$

$$T = 19 + 273 = 292 \text{ K}$$

$$pV = nRT$$

$$150000(V) = (0.5)(8.31)(292)$$

$$V = \frac{(0.5)(8.31)(292)}{150000} = 8.09 \times 10^{-3} \text{ dm}^3$$
$$= \boxed{8.09 \text{ m}^3} \checkmark$$

A flask of 2 dm^3 contains 5.28g of gas. Pressure is 200 kPa & temp. is 20°C . Calculate M_r

$$p = 200 \text{ kPa} = 200000 \text{ Pa}$$

$$V = 2 \text{ dm}^3 = 0.002 \text{ m}^3$$

$$n = ?$$

$$R = 8.31$$

$$T = 20^\circ\text{C} + 273 = 293 \text{ K}$$

$$pV = nRT$$

$$n = \frac{m}{M_r}$$

$$pV = \frac{m}{M_r} RT$$

$$M_r = \frac{mRT}{pV} = \frac{(5.28)(8.31)(293)}{(200000)(0.002)}$$

$$M_r = 32.14$$

$$= \boxed{32 \text{ g/mol}} \checkmark$$

since we don't know the gas, we can't figure the moles out directly

Ionic Lattices

13/12/24

• properties

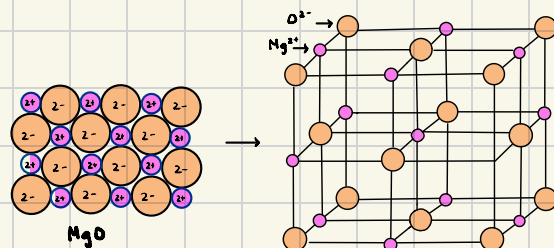
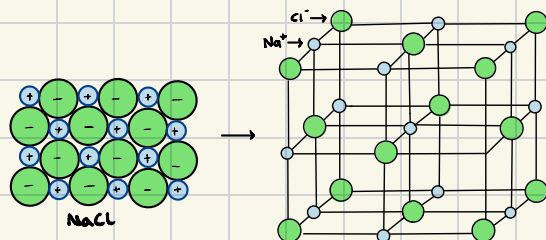
→ structure: hard & brittle

→ melting/boiling point: high, a lot of energy is needed to overcome the strong electrostatic forces of attraction

→ solubility: most are soluble due to the ability to form ion-dipole bonds

→ electrical conductivity: can only conduct electricity when molten or in solution

e.g



Simple molecular lattices

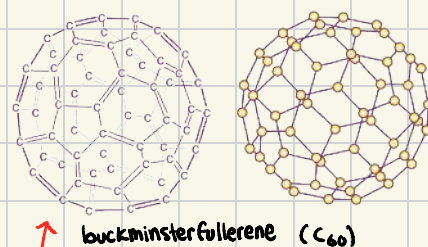
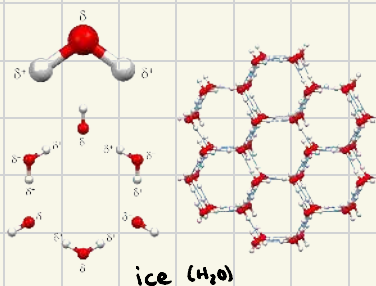
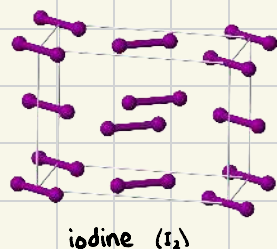
• properties

→ melting/boiling point: low, due to weak intermolecular forces (increases with more electrons)

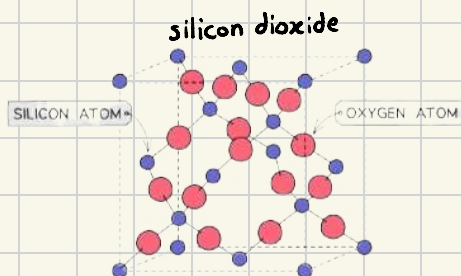
→ solubility: polar substances are soluble in polar solvents & non-polar substances are soluble in non-polar solvents (there are exceptions)

→ electrical conductivity: cannot conduct electricity due to absence of mobile charged particles

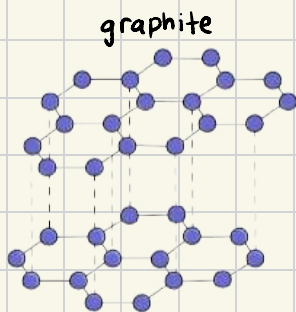
e.g



Giant molecular lattices

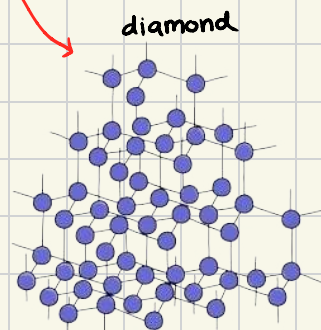


- 🔨 high strength
- 🔥 high melting point (1710°C)
- ⚡ doesn't conduct electricity
- 🚰 insoluble



- 🔨 low strength
- 🔥 high melting point (3600°C)
- ⚡ does conduct electricity
- 🚰 insoluble

allotropes: different structural forms of the same element



- 🔨 high strength
- 🔥 high melting point (4000°C)
- ⚡ doesn't conduct electricity
- 🚰 insoluble

Giant metallic lattices

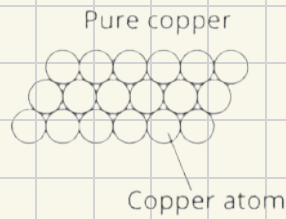
• properties

- melting/boiling point: all transition metals have high MP/BP, some metals have low MP/BP
- solubility: generally insoluble
- electrical conductivity: very good electrical conductors due to 'sea' of delocalised electrons

→ malleable

→ ductile

→ superior electrical conductivity



alloys

a mixture of two or more metals
or a metal with a non-metal.
stronger than pure metals.

